



Iron Deficiency Anemia

By [Gloria F. Gerber](#), MD, Johns Hopkins School of Medicine, Division of Hematology
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Iron deficiency anemia results from low or depleted stores of iron, which is needed to produce red blood cells.

- Excessive bleeding is the most common cause.
- People may be weak, short of breath, and pale.
- Blood tests can detect low levels of iron.
- Iron supplements are used to restore iron levels.

Iron deficiency anemia usually develops slowly because it may take several months for the body's iron reserves to be used up. As the iron reserves are decreasing, the bone marrow gradually produces fewer red blood cells. When the reserves are depleted, the red blood cells are not only fewer in number but also abnormally small and pale.

Iron deficiency is one of the most common causes of anemia, and blood loss is the most common cause of iron deficiency in adults. In males and in females who have stopped having menstrual periods, iron deficiency usually indicates bleeding in the digestive tract. In females who have menstrual periods, menstrual bleeding is the most common cause of iron deficiency. [Iron deficiency](#) may also result from too little iron in the diet during times of increased iron requirements such as in infants, young children, adolescent females, and pregnant people. Decreased iron absorption in the digestive tract, called [malabsorption](#), can result from a variety of disorders, of which [celiac disease](#) is the most common.

Did You Know...

- In the United States, anemia rarely results from consuming too little iron because supplemental iron is added to many foods.

Symptoms of Iron Deficiency Anemia

Symptoms of iron deficiency anemia tend to develop gradually and are similar to symptoms caused by other types of anemia. Such symptoms include fatigue, weakness, and paleness.

Many people with severe iron deficiency anemia have [pica](#). People with pica have a craving to ingest something, most commonly ice but sometimes a substance that is not food, such as dirt, clay, or chalk.

Diagnosis of Iron Deficiency Anemia

- Blood tests

Once blood tests show a person has anemia, tests for iron deficiency are often done. In people with iron deficiency, the red blood cells tend to be small and pale. Blood levels of iron and transferrin (the protein that carries iron when it is not inside red blood cells) are measured and compared.

LAB TEST

[Ferritin Blood Test](#)



LAB TEST

[Iron Tests](#)



The most accurate test for iron deficiency is a measurement of the blood level of ferritin (a protein that stores iron). A low level of ferritin indicates iron deficiency. However, sometimes ferritin levels are misleading because they can be falsely elevated (and thus appear normal) due to liver damage, inflammation, infection, or cancer.

Treatment of Iron Deficiency Anemia

- Stopping bleeding
- Iron supplements

Because excessive bleeding is the most common cause of iron deficiency, the first step is to locate its source and stop the bleeding.

Normal dietary iron intake usually cannot compensate for iron loss due to chronic bleeding, and the body has a very small iron reserve. Consequently, lost iron must be replaced by taking iron supplements.

Correcting iron deficiency anemia with iron supplements usually takes 3 to 6 weeks, even after the bleeding has stopped. Iron supplements are usually taken by mouth. Iron supplements make the stool look dark or black and often cause constipation.

An iron supplement is absorbed best with least side effects when taken every other day. Iron should not be taken with food. A source of vitamin C (either orange juice or a vitamin C supplement) may improve

absorption. Dairy foods or calcium supplements block iron absorption and should be taken separately from iron pills.

Iron supplements are typically continued for 6 months after the blood counts return to normal to fully replenish the body's reserves. Sometimes, iron is given by vein (intravenously) when large amounts of iron are needed or when the person cannot tolerate taking iron by mouth.

Blood tests are done periodically to ensure that the iron supply is sufficient.

Treating the iron deficiency treats pica.



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